Yifan Yang

Research Assistant | M.Sc. in Artificial Intelligence and Robotics
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Summary

Research assistant advised by Pinjia He and M.Sc. student working at the intersection of AI for Software Engineering and AIOps, with a focus on microservice diagnostics, observability, and ML-assisted troubleshooting.

Education

M.Sc. **The Chinese University of Hong Kong, Shenzhen** — Sept 2024 – present
Artificial Intelligence and Robotics — Shenzhen
- School of Data Science.

B.Mgt. **Shandong University** — Sept 2019 – June 2023
Industrial Engineering — Jinan
- Training in systems thinking, optimization, and engineering practice.

Experience

School of Data Science, The Chinese University of Hong Kong, Shenzhen, Research Assistant – Shenzhen — Aug 2024 – present

- Work under the supervision of Pinjia He at the School of Data Science.
- Conduct research in AI for Software Engineering and AIOps, with an emphasis on microservice diagnostics, root cause analysis, observability, and fault-propagation-aware evaluation.
- Contribute to end-to-end research workflows across multiple projects, including system implementation, telemetry and data collection, fault injection, system integration, baseline reproduction, and performance evaluation.
- First author of Gleaner, an online sampling framework for microservice diagnostics accepted at ISSTA 2026.

Publications

Gleaner: A Semantically-Rich and Efficient Online Sampler for Microservice Diagnostics — Apr 2026
Yifan Yang, Aoyang Fang, Songhan Zhang, Pinjia He*
Accepted at ISSTA 2026

Rethinking the Evaluation of Microservice RCA with a Fault Propagation-Aware Benchmark — Oct 2025
Aoyang Fang, Songhan Zhang, **Yifan Yang**, Haotong Wu, Junjielong Xu, Xuyang Wang, Rui Wang, Manyi Wang, Qisheng Lu, Pinjia He*
Accepted at FSE 2026

DynaCausal: Dynamic Causality-Aware Root Cause Analysis for Distributed Microservices — Oct 2025
Songhan Zhang, Aoyang Fang, **Yifan Yang**, Ruiyi Cheng, Xiaoying Tang, Pinjia He*
Under peer review (ASE 2026 submission)

Metis: An Interpretable and Unified Troubleshooting Framework for Microservices using Multi-modal Data — 2024
Zhouruixing Zhu, **Yifan Yang**, Aoyang Fang, Yidan Wang, Pinjia He*
Under review at ACM Transactions on Software Engineering and Methodology (TOSEM)

Projects

Gleaner — Diagnosis-oriented online trace sampling for microservice observability.

chaos-experiment — Fault injection tooling for microservice troubleshooting experiments.

Aegis — Experiment tooling and systems engineering workspace for AIOps research.

Skills

Research Areas: AI for Software Engineering, AIOps, microservice diagnostics, root cause analysis, observability
Systems: Linux, Docker, Kubernetes, Helm, Skaffold, OpenTelemetry, Jaeger
Programming: Python, Go, Shell, SQL
ML and LLM: PyTorch, vLLM, LLM fine-tuning, inference optimization
Languages: Chinese (Native), English (IELTS 6.5)